

Ultimate Basic — C64 compiler

A modern BASIC-like language that compiles directly to 6502 machine code for the Commodore 64. Output: `.prg` files (VICE or real hardware) and `.d64` disk images.

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Quick start

```
cargo build --release

# Compile to .prg (memory map printed on success)
ub build demo.ub -o demo.prg

# Verbose: also prints zero-page layout and hex dump
ub build demo.ub -v

# Compile+ create .d64 disk image
ub build demo.ub --d64 disk.d64

# Compile + create .d64 with extra files embedded
ub build demo.ub --d64 disk.d64 --add music.prg --add loader.prg
```

Language reference

Variables and constants

All variables must be declared with `var` before use. Using an undeclared variable in an expression, assignment, `for/loop` counter, `inc/dec`, `input`, or `read` statement produces a compile-time error.

```
var x = 10           # 8-bit integer (default)
var ptr: word = $0400 # 16-bit – two zero-page bytes, usable as 16-bit
                      # address
var f: float = 3.5    # Q8.8 fixed-point – hi byte = integer, lo byte =
                      # fraction
var msg = "HELLO"     # string variable (pointer to inline PETSCII data)
var s: string = "TEXT" # string with explicit type
var scores = array(10) # byte array, 10 elements stored at $C000+
var times = array_word(8) # word array, 8 word elements stored at $C000+
const BORDER_ADDR = $D020 # compile-time constant (substituted inline, no ZP
                          # slot)
```

Keywords and identifiers are **case-insensitive**: `PRINT`, `Print`, and `print` are all valid.

Type	Width	Notes
int	8-bit	default for numeric literals
word	16-bit	two ZP bytes; can be used as address in poke/peek
float	16-bit Q8.8	hi byte = integer part (0–255), lo byte = fractional part
string	pointer	ZP pair → null-terminated PETSCII in code segment
array(N)	N bytes	byte elements; lives at \$C000+, not in ZP
array_word(N)	Nx2 bytes	word (16-bit) elements; lives at \$C000+, not in ZP

Comments

```
# hash comment
rem this is also a comment
var x = 5 # inline comment
var x = 5 : var y = 6 # colon separates statements on one line
```

Operators

```
x = x + 1
y = a * b - c / 2
z = x and 15          # bitwise AND
w = a or b            # bitwise OR
v = a xor b           # bitwise XOR
m = x shl 3           # shift left
n = x shr 2           # shift right
r = x mod 40          # 8-bit modulo (remainder); SEC/SBC/BCS loop
b = bnot x            # bitwise NOT: x XOR 255 (complement all 8 bits)
```

Comparisons: == != < > <= >= (return 1/0)

Increment / Decrement

```
inc x          # x = x + 1 (INC zp – single instruction)
dec x          # x = x - 1 (DEC zp – single instruction)
```

For **word** variables carry is handled: **inc** uses **INC lo**; **BNE skip**; **INC hi**; **dec** uses **LDA lo**; **BNE skip**; **DEC hi**; **DEC lo**.

Compound Assignments

```
x += 5          # x = x + 5
x -= 3          # x = x - 3
x *= 2          # x = x * 2
x /= 4          # x = x / 4
```

```

x and= 15          # x = x and 15    (bitwise AND)
x or= 64           # x = x or 64     (bitwise OR)
x xor= 255        # x = x xor 255   (bitwise XOR)
x shl= 2          # x = x shl 2
x shr= 1          # x = x shr 1

```

Print

```

print "HELLO"
print x
print x, y, "text"
print                # blank line

print spc(5)         # print 5 space characters
print tab(20), "VALUE" # move cursor to column 20, then print
print "A", spc(3), "B" # mix freely

print "A=" + a       # string + numeric var
print s1 + s2        # two string vars → sequential print
print "Hello " + "World" # two literals → compile-time fold
print chr$(13)       # print by PETSCII code

```

chr\$

```

print chr$(65)      # output character with PETSCII code 65 ('A')
var c = chr$(n)     # store byte value n in variable c
print ">" + chr$(42) # usable in string concat

```

Branching

```

if x == 1 then
  print "YES"
else
  print "NO"
end

```

Select / Case

```

select x
case 1:
  print "ONE"
case 2:
  print "TWO"
else:
  print "OTHER"
end

```

`select` `expr` evaluates the expression once and compares it against each `case` value in order. The first matching case body is executed and control jumps to after `end`. The optional `else:` body runs if no case matches. All values must be 8-bit (0–255).

Loops

```
loop 5                # counted loop (5 iterations)
  print "HI"
end

times 5               # alias for loop N ... end
  print "HI"
end

loop                  # infinite loop
  x = x + 1
  if x == 5 then continue end # skip to next iteration
  if x == 100 then break end
end

var i = 0
for i = 1 to 10        # for..next (preferred)
  if i == 5 then continue end # skip to increment step
  print i
next

for i = 0 to 20 step 2
  print i
next i                 # variable name after 'next' is optional

var i = 0
loop i = 1 to 10        # legacy loop..end syntax – identical code
  print i
end

while x < 100
  x = x + 1
end

repeat                # do-while: body runs at least once
  x = x + 1
until x == 100         # exits when condition is true
```

Labels and goto

```
label main_loop
  x = x + 1
  if x < 10 then goto main_loop end
```

Forward `goto` (label defined later) is fully supported.

`gosub label / return` jump to a label and return back (JSR / RTS at machine code level). The label must be a `label name` statement, not a `sub` — it has no parameters and shares the same zero-page scope. `gosub` supports forward references (label defined after `gosub`).

```
gosub draw_border
...
label draw_border
  # ... draw something ...
  return           # RTS – returns to the instruction after gosub
```

Subroutines

```
sub greet()
  print "HELLO!"
end

sub set_color(col)
  color border col
  color text  col
end

greet()           # call with parens
call greet        # call keyword (no parens)
set_color(6)
```

Parameters are passed via dedicated zero-page slots. No recursion (slots are static). Typed parameters are supported: `sub draw(x, y:int)` or `sub copy(src:string)` — string parameters receive a 2-byte pointer so the callee can index the source string via `src[i]`.

Functions (return values)

```
fn square(x)
  return x * x
end

fn add(a, b)
  return a + b
end

fn clamp(val, lo, hi)
  if val < lo then return lo end
  if val > hi then return hi end
  return val
end

var s = square(9)      # s = 81  – fn call as expression
var c = clamp(joy, 0, 39) # c = clamped value
print add(10, 20)      # 30  – usable inline in print
```

Functions support an optional : **word** return type for 16-bit values:

```
fn get_addr(): word
    return $C000
end

var ptr: word = get_addr() # ptr = $C000
poke ptr, 42               # STA (ptr),Y – valid indirect addressing
var v = peek(ptr)          # LDA (ptr),Y
```

fn is emitted in pass 2 (same as **sub**), so function bodies are never executed at startup.

Forward references are fully supported.

Arrays

```
var scores = array(8)      # 8 bytes at $C000

scores[0] = 100             # constant index → STA $C000
scores[i] = 99              # variable index → STA (ptr),Y
var v = scores[i]          # LDA (ptr),Y
print scores[2]            # usable inline in print

var times = array_word(8)  # 16 bytes (8×2) at $C000+

times[0] = $1234           # constant index → STA $C000 (lo), STA $C001 (hi)
times[i] = $5678           # variable index → ASL A for stride; (ptr),Y × 2
var t: word = times[1]     # LDA $C002, LDA $C003
```

16-bit (word) variables

```
var ptr: word = $0400      # two ZP bytes: lo=$00 hi=$04
poke ptr, 6                # STA (ptr),Y
var v = peek(ptr)          # LDA (ptr),Y
```

Bitmap graphics

```
graphics on                # VIC-II hires bitmap mode (320×200, 1bpp); bitmap
at $2000                  #
graphics on multi          # VIC-II multicolor bitmap mode (160×200, 2bpp, 4
colours/cell)             #
graphics off               # return to text mode

gcls                      # clear bitmap (fills $2000–$3FFF) + set video
matrix colors              #

plot x, y                  # set pixel at (x, y); x: 0–319, y: 0–199
plot erase x, y            # clear pixel (AND ~mask)
plot xor x, y              # toggle pixel (EOR mask) – flicker-free animation
```

```

circle x, y, r          # midpoint circle centered at (x, y) with radius r;
clips off-screen points
line x1, y1, x2, y2     # Bresenham line from (x1,y1) to (x2,y2); x: 0-319,
y: 0-199
line erase x1, y1, x2, y2 # clear pixels along the line (AND ~mask)
line xor x1, y1, x2, y2  # toggle pixels along the line (EOR mask)
rect x1, y1, x2, y2     # draw rectangle outline (4 edges); x: 0-319, y: 0-
199
rect erase x1, y1, x2, y2 # clear rectangle outline (AND ~mask)
rect xor x1, y1, x2, y2  # XOR rectangle outline (EOR mask)
paint x, y              # 4-connected flood fill from (x, y); fills clear
pixels bounded by set ones
mplot x, y, color       # set multicolor pixel (x: 0-159, y: 0-199, color:
0-3); requires graphics on multi

```

Both `graphics on` variants blank the display (LDA \$D011 / AND #\$EF / STA \$D011) while switching VIC registers, then re-enable it in the target mode — prevents mode-switch glitches.

Block graphics (80×50)

```

graphics on block       # 80×50 block-pixel mode (text mode + custom 4-
pixel charset @ $2800)
graphics off            # return to text mode
gcls                    # clear block playfield: screen RAM $0400-$07FF +
color RAM $D800-$DBFF

plot4 x, y              # set block pixel at (x, y); x: 0-79, y: 0-49
plot4 erase x, y        # clear block pixel at (x, y)
circle4 x, y, r         # draw midpoint circle in block pixels; clips to
80×50

```

A chunky low-res mode layered on standard 40×25 text. A 16-character custom charset copied to \$2800 encodes a 2×2 quadrant grid per character (bit3=TL, bit2=TR, bit1=BL, bit0=BR), so each text cell holds 2×2 block pixels → an effective 80×50 grid. No bitmap RAM is used (\$2000-\$3FFF stays free), making it faster than hires bitmap. `plot4` OR's the quadrant bit into the cell so overlapping pixels accumulate; `plot4 erase` clears it. `circle4` uses the same block-pixel helper to draw an outline circle in the 80×50 coordinate space. `gcls` clears both screen and color RAM. See `examples/block_demo.ub`.

Screen and color

```

cls                    # clear screen (KERNAL $E544)
cls fast              # fast fill: screen RAM + color RAM + HOME

color text 14         # text color register $0286
color border 6        # $D020
color bg 0            # $D021

screen 0, 0, 65       # write char 65 ('A') to screen RAM at col 0, row 0

```

```

($0400)
screen 10, 5, ch          # col 10, row 5 – col/row can be variables
screen 5, 3, 42, 7        # char 42 at col 5, row 3, color 7 (writes color
RAM $D800 too)
screen x, y, ch, col      # all four arguments as variables

display on                # re-enable VIC display ($D011 DEN bit)
display off               # blank display

lowercase                 # CHR$(14) → switch VIC-II to lowercase/upercase
charset
uppercase                 # CHR$(142) → switch VIC-II back to
uppercase/graphics charset

cursor 20, 10             # move cursor to col 20, row 10 (KERNAL PLOT $FFF0)
cursor x, y               # column from variable x (0–39), row from y (0–24)

print at 20, 10, "HELLO" # cursor(20,10) + print in one statement (no
trailing newline)
print at x, y, "Score:", score # any mix of exprs
print at 0, 0             # position only (no text, no newline)

scroll x 3                # set horizontal fine scroll: $D016 bits 0–2 = 3
(0–7)
scroll y 2                # set vertical fine scroll: $D011 bits 0–2 = 2
(0–7)
scroll x n                # value can be a variable or expression (masked to
bits 0–2)
scroll x 7 narrow         # set fine scroll and force 38-column mode (hide
edge column)
scroll x 0 wide           # set fine scroll and restore 40-column mode
scroll row 12 left        # shift screen RAM row 12 left by one character

```

`lowercase` emits `LDA #$0E; JSR $FFD2` at runtime. String literals compiled after `lowercase` have their case swapped automatically — uppercase source chars are stored in the PETSCII lowercase slot (\$61+) and lowercase source chars in the uppercase slot (\$41+) — so "Hello World" source displays as **Hello World** on screen. `uppercase` emits `LDA #$8E; JSR $FFD2` and reverts to direct mapping. `cls` does **not** reset the charset mode.

`scroll x n` writes `(n AND 7)` into bits 0–2 of `$D016` (preserving bits 3–7).

`scroll x n narrow` writes the fine-scroll bits and clears `$D016` bit 3 (38-column mode).

`scroll x n wide` writes the fine-scroll bits and sets `$D016` bit 3 (40-column mode).

`scroll y n` writes `(n AND 7)` into bits 0–2 of `$D011` (preserving bits 3–7).

`scroll row R left` shifts one constant screen row left; write the new rightmost character with `screen 39, R, ch`.

Useful for smooth hardware scrolling: decrement each frame from 7 down to 0, shift screen RAM, reset to 7.

`screen col, row, char [, color]` writes directly to screen RAM (`$0400 + row*40 + col`) and optionally to color RAM (`$D800 + row*40 + col`). Constant col/row: address computed at compile time.

Ultimate 64 — CPU Speed

```
speed 4          # set CPU to 4 MHz (reads $D031, updates bits 0-3,
writes back)
speed 48         # 48 MHz (maximum speed on U64)
speed max        # same as speed 48 (alias)
speed off        # back to 1 MHz (alias for speed 1)

badlines on      # enable badline timing ($D031 bit 7 = 0, default C64
behaviour)
badlines off     # disable badline timing ($D031 bit 7 = 1, more CPU
cycles)

var t = turbo()  # 1 if turbo is active (bits 0-3 of $D031 != 0), 0 if
at 1 MHz
```

Available MHz values: 1, 2, 3, 4, 5, 6, 8, 10, 12, 14, 16, 20, 24, 32, 40, 48.

Constant values are rounded down to the nearest available speed at compile time.

Variable values are treated as a raw speed index (0–15) and OR'd into bits 0–3 of \$D031.

\$D031 index	MHz (U64)	MHz (U64 Elite-II)
0	1	1
3	4	4
6	8	10
11	20	24
15	48	64

Requires **U64 Turbo Control** set to **U64 Turbo Registers** or **Turbo Enable Bit** in the U64 config menu.

On a regular C64 or emulator without the register, the poke to \$D031 is silently ignored.

Keyboard

```
var key = getch()      # busy-wait on $FFE4 until key; returns PETSCII
code
var k    = inkey()      # non-blocking: returns PETSCII code, or 0 if no
key pressed
waitkey              # wait until any key is pressed (CIA1 matrix scan;
works during IRQ)
var k    = waitkey()    # same but returns raw $DC01 column bits (0 bit =
key pressed)
var j = joy(2)          # read joystick port 2; returns inverted bits 0-4
var j = joy(1)          # read joystick port 1
                        # bit0=up(1) bit1=down(2) bit2=left(4)
bit3=right(8) bit4=fire(16)
var mx = mouse_x()      # 1351 mouse: accumulated X position (helper:
charge+delay+delta)
```

```

var my = mouse_y()      # 1351 mouse: accumulated Y position (EOR #$FF
inverted)
var mb = mouse_btn()    # mouse buttons: reads $DC01 – bit0=left (fire),
bit1=right (up pin)

```

`mouse_x()` / `mouse_y()` internally run a helper subroutine that handles:

CIA1 \$DC00 capacitor charging, ~516-cycle delay, SID POT X/Y register read (\$D419/\$D41A), signed 7-bit delta computation, Y-axis EOR #\$FF inversion, and accumulated position tracking (permanent ZP state, 8 bytes). The helper samples POT twice per frame. `mouse_y()` does **not** call the helper (only `mouse_x()` does) — it just reads the cached `accum_y` to avoid double-update.

The helper maintains a 9-bit X accumulator (`accum_x` + `accum_x_hi` with carry/borrow). Use `var sx: word = mouse_x()` — the compiler stores both lo and hi bytes, so the `sprite` command correctly handles \$D010 for X positions beyond 255.

The helper stores its state in ZP (\$02–\$09). The init flag must be zero before the first call — zero the area with `fill $02, 7, 0` early in your program. Also disable CIA1 interrupts (`poke $DC0D, $7F`) to prevent the KERNAL IRQ from touching \$DC00 during keyboard scanning — that disturbs the POT reading.

See `examples/mouse_demo.ub` for a working 1351 mouse demo with sprite tracking and button feedback.

Exit

```

bye                # JSR $E544 (clear screen), clear STOP flag, RTS to
BASIC
exit               # alias for bye

```

Timing

```

wait 50            # wait 50 raster-line transitions (~3.2 ms)
wait raster 100    # spin until $D012 == 100 (raster-split effects)
delay 1            # wait 1 PAL frame (1/50 s ≈ 20 ms)
delay 20           # wait 20 frames ≈ 0.4 s; n can be a variable (0–
255)

```

`delay N` counts N complete PAL frames using raster line 200 as the frame boundary.

SID Sound

```

sound 0, $1CAD, 25  # voice 0, freq $1CAD (≈ middle C PAL), 25 frames
duration
sound 1, freq_word, 50 # voice 1, freq from word var, 50 frames (1 s at 50
Hz)

```

```

sound 2, 0, 0          # voice 2, silence

sid volume 15          # master volume full ($D418 = $0F); range 0-15
sid volume 0           # silence (master volume = 0)
sid stop               # zero all 25 SID registers ($D400-$D418) –
complete silence

```

sound <channel>, <freq>, <duration> — duration in PAL frames (1/50 s each).

Fixed ADSR: attack/decay \$09, sustain/release \$F0, sawtooth waveform.

Master volume \$D418 always set to \$0F.

sid volume N writes N to \$D418. Bits 0-3 = volume (0-15), bits 4-7 = filter mode.

sid stop emits a 10-byte zero-fill loop — faster than 25 individual pokes.

Music playback

music play/stop/pause/resume is a high-level alternative to the manual sys sid_init / cia_timer setup.

Requires a prior load sid statement (defines sid_init / sid_play).

```

load sid "tune.sid"    # embed SID file (defines sid_init / sid_play)

music play             # initialise sub-tune 0 + start CIA1 50 Hz IRQ
music play 1           # start from sub-tune 1 (song number 0-based)
music stop             # stop playback + zero all 25 SID registers
($D400-$D418)
music pause            # disable CIA1 timer A IRQ (music freezes, SID
unchanged)
music resume           # re-enable CIA1 timer A IRQ (continues from
pause point)

```

Statement	Effect
music play [n]	call sid_init(n), configure CIA1 timer A at 19 656 cycles (~50 Hz PAL), install IRQ wrapper
music stop	disable CIA1 IRQ + zero all 25 SID registers
music pause	disable CIA1 IRQ (SID output remains frozen)
music resume	re-enable CIA1 IRQ (continues from pause point)

The IRQ wrapper (emitted once at the end of the program) does: ACK CIA1 timer A → JSR sid_play → JMP \$EA81.

Sprites

```

sprite 0, x, y, $2000  # sprite 0: set X, Y position and data pointer
sprite 0, x, y         # without data pointer (keeps existing)
sprite on 0            # enable sprite 0 ($D015 |= bit0)

```

```

sprite off 0                # disable sprite 0 ($D015 &= ~bit0)
sprite color 0, 7           # sprite 0 color = yellow ($D027)
sprite multicolor 0, on     # enable multicolor mode for sprite 0 ($D01C |=
bit0)
sprite multicolor 0, off    # disable multicolor mode ($D01C &= ~bit0)
sprite expand x 0, on       # double width ($D01D |= bit0)
sprite expand x 0, off      # normal width ($D01D &= ~bit0)
sprite expand y 0, on       # double height ($D017 |= bit0)
sprite expand y 0, off      # normal height ($D017 &= ~bit0)
sprite priority 0, on       # behind background ($D01B |= bit0)
sprite priority 0, off      # in front of background ($D01B &= ~bit0)
var h = sprite_hit()        # sprite-sprite collision ($D01E, cleared on read)
var b = sprite_bg_hit()     # sprite-background collision ($D01F, cleared on
read)
var x = sprite_x(0)         # read sprite 0 X position (lo byte, $D000)
var y = sprite_y(0)         # read sprite 0 Y position ($D001)
sprite_frame 0, $2000       # set sprite 0 data pointer only: $07F8+id =
$2000>>6
                             # (does NOT change X/Y position)

```

X supports full 9-bit range (0–319): use a **word** variable for runtime values > 255.

Sprite data pointer: **data_addr** must be 64-byte aligned; stored as **addr >> 6** at **\$07F8+id**.

Sprite definition

```

sprdef 0
  $00,$3C,$00,  $00,$FF,$00,  $03,$FF,$C0,  $07,$FF,$E0,
  $0F,$FF,$F0,  $0F,$FF,$F0,  $1F,$FF,$F8,  $1F,$FF,$F8,
  $1F,$FF,$F8,  $0F,$FF,$F0,  $0F,$FF,$F0,  $07,$FF,$E0,
  $03,$FF,$C0,  $00,$FF,$00,  $00,$3C,$00,  $00,$00,$00,
  $00,$00,$00,  $00,$00,$00,  $00,$00,$00,  $00,$00,$00,
  $00,$00,$00
end

```

sprdef id ... end embeds 63 sprite bytes at the next 64-byte-aligned address in the code segment, emits a **JMP** over them, and automatically sets **\$07F8+id = data_addr >> 6**.

To use the same shape for multiple sprites, read back the pointer:

```

var pg = peek($07F8)      # pointer set by sprdef 0
poke $07F9, pg            # copy to sprites 1–7

```

Custom charset

```

charset $3800              # set base address for chardef (default $3800)

chardef 65                 # redefine character 65 ('A')
  $18,$3C,$66,$7E,$66,$66,$66,$00
end

```

```
chardef 66                # fewer than 8 bytes are zero-padded
    $7C,$66,$7C,$66,$7C
end
```

`charset base` sets the destination address used by all subsequent `chardef` statements. `chardef id ... end` embeds 8 bytes inline in the code segment (preceded by a `JMP` to skip over them), then copies them to `charset_base + id*8` at runtime. Values must be compile-time constants; use `%` for binary literals (`%00011000`).

To activate a custom charset in VIC-II, set the character generator address via `$D018`:

```
charset $3800
chardef 1  $FF,$81,$81,$81,$81,$81,$81,$FF end # box border
poke $D018, $1A    # screen at $0400, charset at $3800 (bank 0)
```

Memory

```
poke $D020, 2          # STA $D020
poke addr_var, 6        # STA (addr_var),Y - if addr_var is word type
var v = peek($D012)     # LDA $D012
var v = peek(addr_var)  # LDA (addr_var),Y - if addr_var is word type

var w: word = peek16($C000) # read 16-bit little-endian: lo=$C000,
hi=$C001
poke16 $0314, $EA81      # write 16-bit little-endian: lo→$0314,
hi→$0315
poke16 ptr, w            # word var as address; word var as value
```

`peek16(addr)` reads two consecutive bytes (lo, hi) as a *word*. `poke16` writes lo then hi.

Disk I/O

```
load "PROGRAM"          # KERNAL LOAD: loads file from device 8 to its
native address
load "DATA", $C000       # loads file to a specific address
load "DATA", ptr         # addr from word variable

save "DATA", $C000, 4096 # KERNAL SAVE from $C000, 4096 bytes → device 8
save "PROG", start, len  # addr and len from word/int variables
```

`load` calls KERNAL `SETNAM+SETLFS+LOAD` (`$FFBD/$FFBA/$FFD5`).

Without address: secondary address 0 (file's own 2-byte header used as load address).

With address: secondary address 1 (file loaded to specified location).

`save` calls `SETNAM+SETLFS+SAVE` (`$FFBD/$FFBA/$FFD8`). Requires both `addr` and `len`.

SID Music

```
load sid "tune.sid"           # embed SID music at its native load address
load sid "tune.sid", $2000    # override: embed at $2000 regardless of SID
header
```

`load sid` reads a PSID or RSID file at **compile time**, strips the header, and appends the raw music bytes to the output `.prg`. After `load sid`, two compile-time constants become available:

Constant	Description
<code>sid_init</code>	Init routine address — call once with A = song number (0-based)
<code>sid_play</code>	Play routine address — call every frame (50 Hz PAL) from an IRQ handler

Both constants work anywhere a constant address is accepted: `sys`, `irq`, `poke`, expressions.

Typical usage:

```
load sid "music.sid"

sub music_irq()
    poke $D019, $FF    # ACK VIC raster IRQ
    sys sid_play        # advance one frame of music
    irq_exit            # JMP $EA81: restore A/X/Y + RTI (proper IRQ exit)
end

sys sid_init, 0        # initialise SID chip: A=0 → first sub-tune
irq music_irq, $C0     # raster IRQ at line $C0 → 50 Hz on PAL

sid volume 15          # master volume on
```

Notes:

- SID data is placed **after** all generated code, padded with zeros up to the load address. The compiler warns if the SID load address would overlap generated code.
- PSID v1 and v2 are supported. If the SID header's load address is 0, the first two data bytes are used as the address (PRG-style, little-endian).
- Only one `load sid` per program is meaningful (the last one wins).

Serial channel file I/O

```
open 1, 8, 2, "MYFILE"    # open logical file 1, device 8, secondary 2, name
                           "MYFILE"
open 2, 4, 7              # open printer (device 4), no filename
open ch, dev, sec         # channel, device, secondary from variables

print# 1, "HELLO"         # send "HELLO"+CR to logical file 1
print# ch, x, "text"      # any mix of vars, strings – same as print but to
                           file
```

```
close 1          # close logical file 1
close ch         # channel from variable
```

`open` calls `SETNAM` (\$FFBD) + `SETLFS` (\$FFBA) + `OPEN` (\$FFC0). Without a filename, `SETNAM` is called with length 0.

`print#` routes output via `CHKOUT` (\$FFC9), `CHROUT` per char (+ trailing CR), then `CLRCHN` (\$FFCC).

`close` puts the channel number in A and calls `CLOSE` (\$FFC3).

Input

```
input score      # read up to 3 digits from keyboard → 8-bit int var
input "Name: ", name  # optional prompt string, then read line → string
var
input "Score: ", score # prompt + int input
```

`input` uses `KERNAL BASIN` (\$FFCF) for blocking, echoed line input with DEL support.

- **Int var:** accepts only 0–9, max 3 chars; converts to 8-bit value on CR.
- **String var:** accepts up to 30 chars; stores as null-terminated string; ZP pair updated.

Float / Fixed-Point

`float` variables use Q8.8 fixed-point format: the high byte is the integer part (0–255) and the low byte is the fractional part (0/256 ... 255/256).

```
var f: float = 3.5      # 3.5 → hi=3, lo=128 (= 0x0380)
var g: float = 0         # integer 0 is promoted to 0.0 automatically

f = 1.5                 # Q8.8 literal assignment
f = f + 1.5             # 16-bit Q8.8 arithmetic (result: 3.0)
f = f + g               # float + float

var n = int(f)          # extract integer part (hi byte) → 8-bit int
print f                # prints as "N.DD" (e.g. 3.5 → "3.50", 1.25 →
"1.25")
```

Operation	Example	Notes
Literal	3.5, 0.25, 1.0	parsed as Q8.8 at compile time
Integer promotion	f = 5	stores 5.0 (hi=5, lo=0)
Add/sub	f + 1.5, f - g	16-bit Q8.8 arithmetic
Extract int	int(f)	returns hi byte as 8-bit int
Print	print f	format "N.DD", always 2 fractional digits

Caveat: Arithmetic overflow wraps at 255.255 (no saturation). Multiplication and division of two float variables are not yet supported — use `int()` + integer arithmetic

for those cases.

Math functions

```
var a = abs(x - 20)      # two's-complement absolute value
var b = min(x, 39)       # 8-bit minimum
var c = max(x, 0)        # 8-bit maximum
var s = sgn(score)       # 0 = zero, 1 = positive (1-127), $FF = negative
                        # (128-255)
var r = rnd()            # LCG pseudo-random 0-255; seed from raster line
var r = rnd(10)          # LCG pseudo-random 0-9 (rnd() mod n; result 0..n-1)
var s = sin(angle)       # sine: angle 0-255 (full circle), returns 0-255
                        # (center=128)
var c = cos(angle)       # cosine = sin(angle+64)
var v = clamp(x, 0, 39)  # 8-bit unsigned clamp: result = max(lo, min(hi, val))

print hex(n)             # print as 2-digit uppercase hex
print bin(n)             # print as 8-bit binary string
```

String functions

```
var n = len(msg)         # length of null-terminated string var (0-255)
var c = asc(msg)         # PETSCII code of first character (0 if empty)
var c = asc("A")         # compile-time: constant PETSCII code
var n = val(s)           # runtime: parse decimal PETSCII string → 8-bit int
                        # (e.g. "042" → 42)
var c = msg[i]           # string character at index i: PETSCII code of
msg[i]                   # msg[i]
msg[i] = c               # write PETSCII byte c to string at index i – STA
                        # (ptr),Y
msg[0] = 72              # constant index → LDY #0; STA (ptr),Y
```

Number formatting

```
print hex(n)             # print as 2-digit uppercase hex
print bin(n)             # print as 8-bit binary string
print dec(n, 4)          # right-justified decimal in a field of 4 chars
                        # (e.g. 42 → " 42")
print dec(n, width)      # width can also be a variable
```

`dec(n, width)` pads the number on the left with spaces to fill `width` characters.

If the number has more digits than `width`, it is printed without padding (no truncation).

In non-print contexts `dec(n, w)` evaluates to `n` unchanged (same as `hex/bin`).

REU (RAM Expansion Unit)


```

var ok = reu_present()    # 1 if REU detected, 0 if not (write/read test on
$DF04)
var ok = reudet()        # alias for reu_present()

reu stash c64addr, bank, reu_addr, len # copy C64 → REU
reu fetch c64addr, bank, reu_addr, len # copy REU → C64
reu swap  c64addr, bank, reu_addr, len # swap between C64 and REU

```

`reu_present()` performs a write/read-back test on REU register `$DF04`. Without an REU the write is lost (open bus), so the read-back differs — reliably detects presence without touching any side-effecting command register.

Parameter	Width	Notes
<code>c64addr</code>	16-bit	C64 RAM start — constant, <code>word</code> var, or 8-bit expr
<code>bank</code>	8-bit	REU bank number (0–7 for a 512 KB unit)
<code>reu_addr</code>	16-bit	Offset within the REU bank
<code>len</code>	16-bit	Bytes to transfer (0 = 65 536 in REU hardware)

REU registers: `$DF01` command (`$B0` stash / `$B1` fetch / `$B2` swap),
`$DF02–$DF03` C64 addr, `$DF04–$DF05` REU offset, `$DF06` bank, `$DF07–$DF08` length.
Transfer is synchronous (CPU halted during DMA).
Requires a real REU or VICE: **Settings → Hardware → RAM Expansion Module**.

Memory utilities

```

fill screen 32          # fill screen RAM $0400–$07FF with value 32 (space
char)
fill color 1            # fill color RAM $D800–$DBFF with value 1 (white)

fill $0400, 1000, 32    # fill 1000 bytes starting at $0400 with value 32
fill addr, 256, 0       # addr can be word var
fill ptr, len_word, val # all operands can be expressions / word vars

memcpy $C000, $0400, 256 # copy 256 bytes from $C000 → $0400
memcpy src_ptr, dst_ptr, 40 # word vars for source and destination

drawmem $C000, $0400, 8, 10, 40 # blit 8×10 rect from $C000 → screen at
$0400, stride 40
drawmem src_ptr, dst_ptr, w, h, 40 # word vars for src/dst

```

Both `fill` and `memcpy` support 16-bit lengths (0–65535). Use `word` variables for lengths > 255.

`drawmem src, dst, width, height, stride` copies a 2-D rectangular block. `src` is read linearly (packed rows); `dst` advances by `stride` bytes between rows — use 40 (\$28) for the C64 screen or color RAM (40 columns). Width, height and stride are all 8-bit values. `src` and `dst` may be constants, `word` variables, or 8-bit expressions.

Raster IRQ

```
irq my_handler          # raster IRQ at line 0, handler = sub name or
address
irq my_handler, 100     # raster IRQ at raster line 100
irq $C800, 200         # handler at fixed address
irq addr_word          # handler address from a word variable
```

Sets up a raster IRQ via the BASIC soft vector (\$0314/\$0315): disables CIA1 timer IRQ, ACKs pending VIC IRQ, writes raster line to \$D012, enables VIC raster IRQ (\$D01A=\$01), writes handler address, and re-enables interrupts.

The handler **must** end with `sys $EA81` (KERNAL end-of-IRQ) — plain `RTS` or `RTI` will corrupt the stack. ACK the VIC IRQ first:

```
sub my_handler()
  poke $D019, $FF      # ACK VIC IRQ
  # ... work here ...
  sys $EA81            # JMP to KERNAL end-of-IRQ
end
```

Forward references are supported (`irq my_handler` before the sub is defined).

NMI handler

```
nmi my_nmi              # set NMI vector $0318/$0319 to handler sub or
address

sub my_nmi()
  # ... NMI work here ...
  nmi_exit              # JMP $FE47 — proper NMI exit (restores A/X/Y +
RTI)
end
```

`nmi handler` writes the handler address to the NMI soft vector (\$0318/\$0319). The hardware NMI vector at \$FFFA points to the KERNAL NMI routine which branches through \$0318. The handler **must** end with `nmi_exit` (emits `JMP $FE47`) — using plain `RTI` will corrupt the stack. Forward references supported.

CIA1 timer IRQ

```
cia_timer 19656, my_handler # CIA1 timer A: fires every 19656 cycles (~50
Hz PAL)
cia_timer period, handler   # period can be a variable or expression
```

Sets up CIA1 timer A as a periodic IRQ source via the BASIC soft vector (\$0314/\$0315):

1. SEI — disable interrupts
2. `$DC0D = $7F` — disable all CIA1 IRQs
3. Load 16-bit period lo→`$DC04`, hi→`$DC05`
4. Write handler address to `$0314/$0315`
5. `$DC0D = $81` — enable CIA1 timer A IRQ
6. `$DC0E = $01` — start timer A in continuous mode
7. CLI — re-enable interrupts

The handler must end with `irq_exit` (or `sys $EA81`) and should ACK the CIA1 IRQ:

```
sub my_handler()
    poke $DC0D, $01      # ACK CIA1 timer A IRQ (read also clears it)
    # ... work here ...
    irq_exit             # JMP $EA81: restore A/X/Y + RTI
end
```

PAL timing: clock = 985 248 Hz. Period for 50 Hz = 985 248 / 50 = 19 705 cycles ≈ `$4CC9`. Forward references supported.

Error handling

```
onerr goto err_handler  # set KERNAL I/O error vector ($0300/$0301) to a
label

...

label err_handler
    print "I/O ERROR"
    bye
```

`onerr goto label` writes the label address (lo, hi) to KERNAL locations `$0300` and `$0301`. When a KERNAL I/O error occurs (e.g. a failed `load` or `open`) the KERNAL executes `JMP ($0300)`, which branches to the label. Forward references (label defined after `onerr goto`) are supported.

Compile-time file embedding

```
incbin "sprites.bin"    # embed raw binary bytes at current code position
include "defs.ub"       # inline another .ub source file (lexed+parsed in
place)
```

Data / Read

```
data 1, 2, 3, 255       # constant byte table
read varname            # load next byte into varname (auto-declares if
needed)
```

All **data** values are collected at compile time. A 2-byte ZP pointer is automatically allocated and initialised at program start. Each **read** advances the pointer.

Inline assembly

```
sys $FFD2                # JSR $FFD2
sys $FFD2, 7             # LDA #7 ; JSR $FFD2 (pass byte value in A
register)
irq_exit                 # JMP $EA81 – proper IRQ handler exit (restore
A/X/Y + RTI)
asm $EA, $EA             # inline raw bytes (NOP NOP) – legacy form
asm {
    ; Full 6502 mnemonics and addressing modes
    LDA #$07              ; immediate
    STA $0286             ; absolute
    LDA $50               ; zero-page ($50 ≤ $FF → ZP auto-selected)
    STA $0400,X           ; absolute,X indexed
    LDA ($50),Y           ; (indirect),Y
    LDA ($50,X)           ; (indirect,X)
    JSR $FFD2             ; subroutine call
    JMP $C000             ; absolute jump
    JMP ($FFFC)           ; indirect jump

    CLC
    ADC #1
    SEC
    SBC #1

    TAX                  ; implied / transfer
    ASL A                ; accumulator (also just: ASL)
    LSR A
    ROL
    ROR

    ; Branches – operand is an absolute address; offset is computed
    automatically
    BNE loop              ; forward or backward branch to local label
    BEQ done

loop:
    NOP
done:
    RTS

    ; #<label / #>label – lo / hi byte of a label address
    LDA #<handler
    STA $0314
    LDA #>handler
    STA $0315

    ; * – current assembly location
    JMP *
```

```

; Raw hex bytes (backward-compatible with old asm { $xx ... } syntax)
$EA $EA          ; two NOP bytes
}

```

Addressing modes:

Syntax	Mode	Bytes	Example
(no operand)	Implied	1	NOP, RTS
A	Accumulator	1	ASL A, LSR
#value	Immediate	2	LDA #\$07
\$zz (0–255)	Zero-page	2	LDA \$50
\$zz,X	ZP,X	2	LDA \$50,X
\$zz,Y	ZP,Y	2	LDX \$50,Y
\$xxxx	Absolute	3	LDA \$0400
\$xxxx,X	Absolute,X	3	LDA \$0400,X
\$xxxx,Y	Absolute,Y	3	LDA \$0400,Y
(\$xxxx)	Indirect	3	JMP (\$FFFC)
(\$zz,X)	(Indirect,X)	2	LDA (\$50,X)
(\$zz),Y	(Indirect),Y	2	LDA (\$50),Y
label	Relative	2	BNE label (branches only)

- \$zz (1–2 hex digits, value ≤ 255) selects zero-page if the instruction supports it; otherwise auto-upgrades to absolute. Use \$00xx (4 digits) to force absolute.
- Branch operands are absolute addresses; the relative byte offset is computed automatically.
- Local labels (name:) are scoped to the asm { } block. Forward branches resolved in pass 2.
- #<label / #>label yield the lo / hi byte of a label's address.
- * yields the current instruction address, so JMP * assembles as a self-loop.
- Lines starting with \$, %, or a digit are emitted as raw bytes (backward-compatible).
- Inside asm { }, comments are ; or // to end of line. (# is the immediate prefix, not a comment.)

Mixing asm { } with subroutine parameters

Parameter names are **not accessible** inside asm { } blocks. Use UltimateBasic statements to move values into known locations before the asm { } block:

```

sub set_colors(border_col, bg_col)
  poke $D020, border_col  # UltimateBasic resolves the ZP address
  poke $D021, bg_col
  asm {
    ; values are already in $D020 / $D021
    LDA $D020

```

```
}  
end
```

For routines whose entire body is assembly — especially IRQ handlers that must cross-reference each other — put all handlers in a **single top-level `asm { }` block** in the main program. Labels in the same block share scope, so `irq1` and `irq2` can reference each other freely. See `examples/raster_irq_demo.ub`.

String ↔ integer

```
numstr score, $0340      # writes "042\0" at $0340 (always 3 digits, zero-  
padded)  
var n = str_to_int("42") # compile-time: Expr::Number(42)  
  
print str$(score)        # print 8-bit int as 3-digit decimal string  
("000"-"255")  
print "Score: " + str$(score) # usable in string concat print context  
var s: string = str$(n)    # assign str$() result to a string var  
(shared static buffer)
```

`str$(n)` converts an 8-bit value to a 3-character decimal string (always 3 digits with leading zeros, e.g. `5` → `"005"`, `42` → `"042"`, `255` → `"255"`) followed by a null terminator. The result pointer is stored in a permanent ZP pair allocated at compile time.

Note: `str$(n)` uses a single shared 4-byte static buffer. Calling `str$(n)` again overwrites the previous result. For concurrent display of multiple values, use `numstr` to write to separate absolute addresses.

Examples

File	Description
<code>examples/features.ub</code>	const, label/goto, poke/peek, rnd, math functions
<code>examples/new_features.ub</code>	sub params, arrays, word vars, string vars
<code>examples/bitmap_demo.ub</code>	320×200 bitmap, plot, graphics on/off
<code>examples/block_demo.ub</code>	80×50 block graphics, plot4, circle4, graphics on block
<code>examples/joystick_demo.ub</code>	joystick reading, sprite movement
<code>examples/mux_demo.ub</code>	raster sprite multiplexer (3 windows × 8 sprites = 24)
<code>examples/orbit_demo.ub</code>	24-sprite orbit with pulsating radius and random colors
<code>examples/plasma_demo.ub</code>	plasma-effect bitmap with raster bar border animation
<code>examples/sprite_data.ub</code>	sprdef shape data (included by other demos)
<code>examples/sprite_mux_orbit.ub</code>	24-sprite orbit demo with sprdef + precomputed positions
<code>examples/sprite_orbit_demo.ub</code>	8 hardware sprites in circular orbit via sin/cos table

File	Description
examples/reu_bitmap_demo.ub	REU stash/fetch with bitmap graphics
examples/sid_music_demo.ub	SID music player with raster IRQ and keyboard exit
examples/tenprint.ub	5 TENPRINT maze implementations with menu; demos lowercase charset mode

Architecture

```
.ub source
  → Lexer → Vec<Token>
  → Parser → Vec<Stmt>
  → Codegen → Vec<u8> (raw 6502 machine code)
  → mod.rs → PRG = BASIC SYS stub + machine code
```

Two-pass compilation: Pass 1 = main code, Pass 2 = subroutine bodies (appended after main `RTS`). Forward references (`call`, `goto`, `plot`) are patched at the end.

Zero-page layout: `$02–$4F` permanent (vars, loop counters, sub params), `$50–$7F` scratch (reset per statement), `$FB` RNG seed.

CLI reference

```
ub build <input.ub> [OPTIONS]

-o, --output <file>  Output .prg file (default: <input>.prg)
-v, --verbose         Show zero-page layout and code hex dump
--no-stub             Skip the BASIC SYS stub (code loads at $0801)
--d64 [file]          Also produce a .d64 disk image;
                     without a filename defaults to <output>.d64
--add <file>          Add an extra file to the .d64 disk image;
                     may be repeated for multiple files
-h, --help            Show help
```

`--add` requires `--d64`. The compiled `.ub` program is always the first file on the disk; each `--add` file is appended after it. File names on the disk are derived from the source file stem, uppercased (e.g. `music.prg` → `MUSIC`).

After a successful build the compiler always prints a memory map:

```
demo.ub → demo.prg (386 bytes)
```

```
Load:    $080D – $0989
```

```
Variables (zero page):
```

```
score    ZP:$08    byte
lives    ZP:$0A    byte
```

```
msg      ZP:$0C   string
ptr      ZP:$0E   word
```

Subroutines:

```
greet    $0900
show     $0912
```

Arrays (\$C000+):

```
sprites  $C000   8 bytes
```

With `-v` the output additionally shows the internal ZP allocations and a full hex dump.

Building

```
cargo build --release    # binary: target/release/ub
cargo test                # unit + integration tests
```

Known limitations

Feature	Limitation
Integer arithmetic	8-bit unsigned (0–255); <code>word</code> vars hold 16-bit values
Subroutines	No recursion — ZP parameter slots are statically allocated
String vars	Read-only after init; assignment replaces the pointer, not the data
String concat runtime	<code>s1 + s2</code> prints sequentially — no heap allocation or length tracking
<code>rnd()</code> / <code>rnd(n)</code>	Simple LCG, not cryptographic; period = 256
<code>abs()</code> / <code>sgn()</code> / <code>min()</code> / <code>max()</code>	8-bit values only; <code>abs/sgn</code> treat values as signed (bit 7 = negative → <code>abs</code> two's-complements, <code>sgn</code> returns <code>\$FF</code>); <code>min/max</code> are unsigned (0–255)
<code>plot</code>	Out-of-range pixels are silently clipped ($Y \geq 200$ or $X \geq 320 \rightarrow$ no-op)
<code>mplot</code>	No bounds checking — x must be 0–159, y must be 0–199
<code>rect</code>	No bounds checking — x: 0–319, y: 0–199; $x1 \leq x2$ and $y1 \leq y2$ not enforced (degenerate/inverted rects produce undefined output)
<code>plot4</code>	No bounds checking — x must be 0–79, y must be 0–49 (block mode)
<code>circle4</code>	Clips off-screen block pixels; useful radius is roughly 0–49 in 80×50 block mode
<code>chr\$</code>	No PETSCII↔ASCII mapping — n is passed as-is to CHROUT
<code>music play</code>	Requires <code>load sid</code> ; only one CIA1 wrapper is emitted (last <code>music play</code> wins)
Error reporting	Compile-time only; <code>onerr goto</code> handles KERNAL I/O errors at runtime